

Point A to Point B: How things have changed

Ever wondered what the actual potential of man is? It's difficult to predict anything while one rests in his own timeline. But one peek through the fabric of time and you may realize what glorious future awaits.



Auto Matrix

Auto Matrix is a vehicular traffic management system that monitors automobiles using complex routing algorithms to direct cars through the city roads. It successfully simulates vehicle traffic congestion due to heavy usage or traffic incidents, and the effects on surrounding areas and figures out the congestion free path for a vehicle through the city roads. The algorithm also has heuristic prediction models that account for special occurrences like festivals and transmits the data directly to the cars on the streets using RFID (Radio Frequency Identification)].



Virtual Mechanic:

Using augmented reality, the mechanic receives additional three-dimensional information on the part being repaired. Apart from the real environment, virtually animated components are seen along with instructions on each of the working steps through headphones integrated inside the goggles.

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Hello, it's me Agent 001 and the Year is 2030. Things have changed a lot since I was a young geek. As an ageing gamer, rather than matters such as the choice between HD 6990 and GTX 590 more immediate concerns like getting to the Digit office on time are on my mind. Yet technology is always at hand.

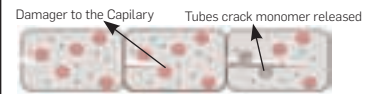
Traffic jams are now a thing of the past and the cursed long waits no longer exist, thanks to the Smart Cars and the **Auto Matrix** that exists today which have taken over the concept of what a car used to be. Today's cars are much more than just cars, they are high tech computers. It really feels good when my car gets itself out of the garage and all ready to go when I come out of my house in the morning, and the best part, I don't have to drive my car, drives itself!

Of course machines will be machines. Hit a snag in the car with no one around to help? No problems, the **virtual mechanic** is there to help you with a highly Interactive Virtual Reality (VR) Interface, it gives you voice and 3D visual help how to get your car back on track. Augmented reality has even revolutionised the old car windshield. Now we get tons of relevant information overlaid on the actual scene. Kind of distracting but certainly useful. Oooh there's an offer at Starbucks!

Advancements in Polymer Technology has lead to the development of **Self Healing Polymers** and photo chromatic paints, something I personally like as it keeps my Hybrid as new as the day I bought it. This technology has also revolutionised the aviation industry.

Present day planes can heal themselves from the wear and tear they get when travelling at high speeds. They have smart seats which use body monitors to sense a tense passenger a calm him down by playing soothing music or ordering a drink.

The good old Railway is next pit stop in the transport exploration of my time. A recent journey on the Delhi-Mumbai High Speed Maglev Corridor took me around three hours. Magnetic Levitation is now an old-tried and tested technology but what powers it is new - Hydrogen fuel cells. Hydrogen for these trains is now developed from human and animal waste using the **Microbial Electrolysis Cell** which makes hydrogen available throughout the country rather than at some distribution/refilling stations.



Self-healing Polymers

These materials repair damage caused by mechanical usage over time. Fragile glass capillaries or fibres are imbedded within a composite material. The resulting porous network is filled with monomer. When damage occurs, the tubes also crack and the monomer is released into the cracks. Other tubes containing a hardening agent also crack and mix with the monomer, causing the crack to be healed.]

Photo chromic polymers

Photo chromic polymers change colours when exposed to sunlight/UV, and reverts to its original colour when the sunlight/UV is dimmed or blocked. After absorbing the energy of sunlight/UV, its molecule structure is changed, which causes its absorbed wavelength to be changed allowing a colour to appear. The change occurs in a second.



Microbial Electrolysis Cell (MEC)

Whilst Microbial Fuel Cells (MFCs) of today produce an electric current from the microbial decomposition of organic compounds, MEC's partially reverse the process to generate hydrogen from organic material by applying an electric current. Electrogenic microorganisms consuming an energy source (such as acetic acid) release electrons and protons, creating voltage.